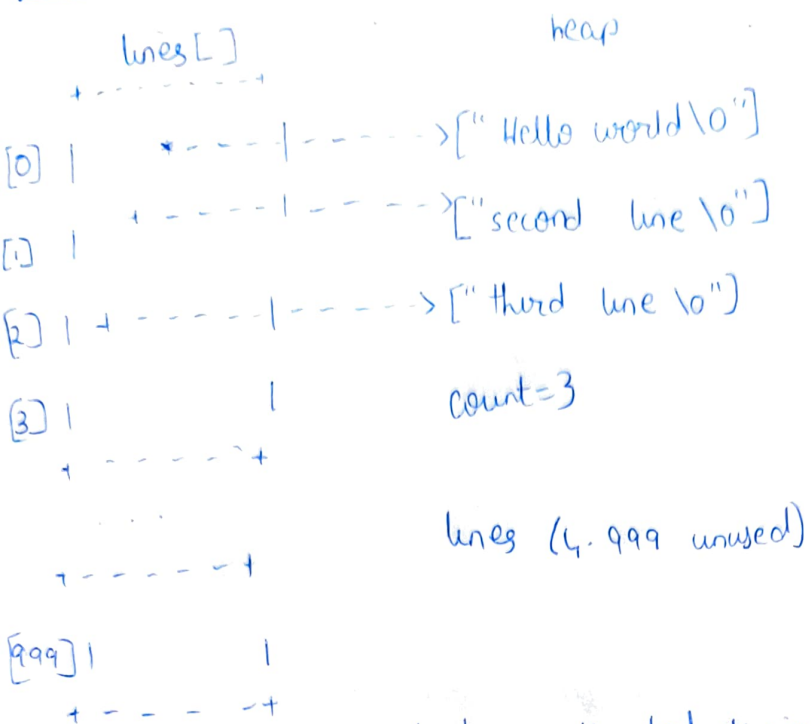


Data Structure :-

→ An array of pointer to heap-allocated strings

Struct :-

```
#define MAX_LINES 1000
#define MAX_LEN 512

typedef struct {
    char* lines[MAX_LINES];
    int count;
```

```
} Document;
```

```
static Document doc; backup;
```

→ We have chosen a structure of fixed array of 1000 pointers each pointing to a malloc'd string

Command Set

command	meaning	example
i<n><text>	Insert text as line n, shift the text down	i 3 hello world
d<n>	Delete line n, shift the text up	d 3
p	print the whole doc with line numbers	p
s<file>	save the document to a file	s notes.txt
l<file>	load a file replacing the doc	l notes.txt
f<word>	find which lines contain a word	f error
r<n><old>	replace first old on line n	r 2 cat dog
h	show help text	h
q	Quit	q

~~Functions List:-~~

~~* helpers *~~

~~void stamp~~

Function List

/* helpers */

void strip(char *s);

int valid(int n);

char *dup_str(const char *s);

void doc_free(Document *d);

void snapshot(void);

/* core */

int cmd_insert(int n, const char *text);

int cmd_delete(int n);

int cmd_print(void);

int cmd_save(const char *f);

int cmd_load(const char *f);

/* bonus */

void cmd_search(const char *word);

int cmd_replace(int n, const char *old, const char *new);

void cmd_status(void);

int cmd_undo(void);

void print_help(void);

④ insert (hand written)

int cmd_insert(int n, const char *text)

{
if (doc.count >= MAXLINE || n < 1 || n > doc.count+1)
return 0;

char *copy = dup_str(text);

if (!copy) return 0;

for (int i = doc.count; i > n-1; i++)

doc.line[i] = doc.line[i-1];

doc.line[n-1] = copy;

doc.count++;

return 1;

}

Draw this beside it - insert "X" at line 3,

count = 3.

before: [A][B][C][C] i=3, copy index 2 → 3

step 2: [A][B][B][C] i=2, copy index 1 → 2

step 3: [A][X][B][C] looks like (i) 1) false

write copy at
index 1

(count+1)

Page (5) - delete 1

incl incl, delete (w n)

```
if (!valid(n)) return 0;
```

```
free (doc->lines[n-1]);
```

```
for (int i = n-1; i < doc->count-1; i++)
```

```
doc->lines[i] = doc->lines[i+1];
```

```
doc->count--;
```

```
return 1;
```

```
}
```

delete line, count = 4;

before: [A][X][B][C] count 4

1 Free this pointer first

Step 1: [A][B][B][C] i = 1, copy index 2 → 1

Step 2: [A][B][C][C] i = 1, copy index 3 → 2

loop stop (i < 3 false)

Step 3: [A][B][C] count 3 - slot 3 is now unused

Page 6:

void _init(void)

{

if (!doc.count) { printf("Document is empty\n");

return;

for (int i = 0; i < doc.count; i++)

printf("%s\n", doc.doc[i]);

}

Edge cases

Situation

iD text -line number 1

999 text on a 3-line doc

14 text on a 3-line doc

15 on a 3-line doc

11 on a 1-line doc

on an empty document

1 missing.txt

5 bad/path.txt

Document already at

1000 lines

closed at the prompt

What the editor does

Reject, error message,
document unchanged

Rejected only count+1 is
valid

Allowed - this is append

Rejected, error message

works, document becomes
empty

prints (doc is empty)

open returns Null

proper - - -

Insert rejected

gets returns Null

, editor exists cleanly

Team

→ Trivikram Kaligoti

→ Sathakam Ram

→ Anand Kumar

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